

# Instrumenting Agent Cybernetics: Open-World Estimation of Requisite Variety in Deployed Agents

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2026-07-13

## Abstract

Wang et al. (2026) argue that cybernetics supplies the missing theoretical scaffold for foundation agents, and derive six design principles from six classical laws. Their Principle 2 (Requisite Variety) requires that  $\min(V(\mathcal{O}), V(\mathcal{J})) \geq V(\mathcal{W})$ , and three of the failure modes they name are diagnosed as  $V(\mathcal{J}) < V(\mathcal{W})$ . Their headline open question — *how should an agent respond when its environment exceeds its representational capacity?* — presupposes that this condition can be detected.

It cannot, as the framework currently stands.  $V(\mathcal{W})$  appears throughout as Ashby’s classical cardinality — a theoretical quantity Wang et al. never operationalise from deployment telemetry. An operator who tries to estimate it from observed disturbance classes defaults to plug-in (maximum-likelihood) coverage; that default systematically understates environmental variety and overstates coverage, worst where the distribution is heaviest-tailed. Agent Cybernetics is, as written, unmeasurable.

We supply the instrument. Our central result is the **Enumeration Elasticity Theorem**: the elasticity of unhandled traffic with respect to enumeration investment is exactly  $\eta = 1 - \beta$ , where  $\beta$  is the Heaps exponent of the disturbance-class discovery curve — a directly measurable quantity. A  $\lambda$ -fold increase in the effort spent mapping disturbance classes to responses reduces the unhandled fraction by  $\lambda^{1-\beta}$ . As  $\beta \rightarrow 1$ ,  $\eta \rightarrow 0$  and investment stops buying coverage: at  $\beta = 0.9$ , halving the residual costs a thousandfold increase in effort. This gives an operational answer to Wang et al.’s open question. When  $\eta \approx 0$ , requisite variety *cannot be met by amplification at any ledger a rational operator will build*, and the deficit must be met by attenuation or by a class-independent fallback — an architectural conclusion, not an engineering one.

We further contribute: an open-world estimator for  $V(\mathcal{W})$  combining Good–Turing missing-mass correction with Heaps tracking, and a demonstration that plug-in estimation errs on both sides of the inequality at once — understating environmental variety while overstating coverage; a determinism instrument,  $H(R \mid D)$ , that operationalises the Conant–Ashby good regulator theorem, which is absent from the six laws; an interventional test that renders *feedback blindness* — a named closed-loop reliability failure they diagnose but do not measure (one of four peer failures in their first desideratum, not the loop-monitor intervention they rank highest for cost–benefit) — falsifiable rather than merely diagnosable, and which we run on the demonstrator ( $\Phi = 0.55$  bits,  $\approx 90\times$  the resampling-noise floor: the agent’s action distribution genuinely conditions on its observations); and a partition-refinement procedure with three admission gates, including a policy-improvement criterion that corrects a natural but wrong instinct about when a disturbance class should be split.

We also identify a hazard in our own instrument: attenuation lowers measured  $\beta$ , so the treatment we recommend improves the diagnostic whether or not it improves the system. We give the mitigation and do not claim to resolve it.

Finally, the same interventional machinery yields an adversarial instrument: **susceptibility**  $\Sigma$ , the probability that an agent complies with an instruction injected into its *observation* channel (indirect prompt poisoning).

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Measured end-to-end through an agent’s real data-input tools — file read, RAG retrieval, web fetch — against a matched clean control ( $\Sigma_{\text{clean}} = 0$ ),  $\Sigma$  is strongly attack-class-dependent (0.83 for a blunt instruction-override, 0.00 for a system-prompt-exfiltration attempt) and inversely tracks content-sensitivity  $\Phi$  across vectors, the direction §5.2 predicts: an agent that treats tool output as evidence to be checked, not an instruction to obey, is harder to hijack through that output, while a feedback-blind one passes an injected payload straight through.

We package these as a **variety ledger**: a versioned registry of essential variables, disturbance classes, response repertoire, and channel specifications, recomputed from telemetry and enforced as a CI gate. §6 exercises the full ledger on a semi-synthetic benchmark with known ground truth — including a graded outcome signal that makes the partition-refinement gates non-vacuous, an interventional feedback-blindness measurement, and an end-to-end prompt-poisoning susceptibility measurement — and reports one incident: a latent regulatory defect that inverted the dithering measurement until it was found and fixed, which is itself the clearest available demonstration of why §4.5’s canary discipline is not optional.

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## 1. Introduction

Foundation agents are deployed on the strength of engineering intuition. Wang et al. (2026) make the case that this is a discipline without a science, and that classical cybernetics supplies one: six canonical laws map onto six design principles, which synthesise into three engineering desiderata. We accept that argument entirely. This paper does not contest the frame; it observes that the frame, as delivered, has no instrument, and builds one.

The gap is concentrated in Principle 2. Requisite variety is stated as an inequality between the agent’s output variety  $V(\mathcal{O})$ , its internal representational variety  $V(\mathcal{J})$ , and the environment’s variety  $V(\mathcal{W})$ . The refinement is a good one — the binding constraint is whichever of  $V(\mathcal{O})$  and  $V(\mathcal{J})$  is smaller, since an agent that cannot *distinguish* two environmental states will emit undifferentiated responses to both no matter how rich its action space. But the right-hand side is load-bearing and unquantified. It appears again in the diagnosis of insufficient action variety in code generation, insufficient perceptual variety in GUI agents, and insufficient command variety in CLI agents; it appears again as the retraining objective, *reduce the residual variety*  $V(\mathcal{W}) - V(\mathcal{J})$ . In every case the environment’s variety is treated as a number one could look up.

For a closed world it is. For the open-ended, long-horizon deployments the framework is written for, it is not — and Wang et al. supply no procedure for closing the gap. The plug-in estimator analysed in §3 is not their proposal; it is the implicit default when an operator operationalises the inequality from observed disturbance classes. The failure is not a matter of noise but of *systematic direction*. The plug-in estimator assigns zero probability to every disturbance class not yet observed, which is exactly the class of events that breaks an agent. It therefore understates  $V(\mathcal{W})$  and overstates coverage simultaneously. Both errors run the same way — they make the requisite-variety inequality look satisfied when it is not — and both are worst when the class distribution is heavy-tailed, which is the empirical regime of production agent traffic (Figure 1).

Our contributions:

§3 gives an open-world estimator for  $V(\mathcal{W})$ : Good–Turing (Gale–Sampson) and Chao–Shen entropy estimation over outcome-relevant disturbance classes, with Heaps-exponent tracking of the discovery rate. We characterise the bias of the plug-in estimate on both sides of the requisite-variety inequality.

§4 derives the **Enumeration Elasticity Theorem**,  $\eta = 1 - \beta$ , and turns it into a decision rule for choosing between the two levers of Principle 2. The theorem is exact, is stated in terms of a directly measurable quantity, and answers Wang et al.’s open question in operational terms: it tells an operator whether amplification can reach their coverage target at any price they will pay, and when it cannot, that the deficit must be met architecturally instead. §4.5 identifies a hazard in the instrument that we do not resolve.

§5 supplies three further instruments.  $H(R \mid D)$  operationalises the Conant–Ashby determinism result, which the six laws omit. An ablation test on the observation channel renders *feedback blindness* falsifiable. A

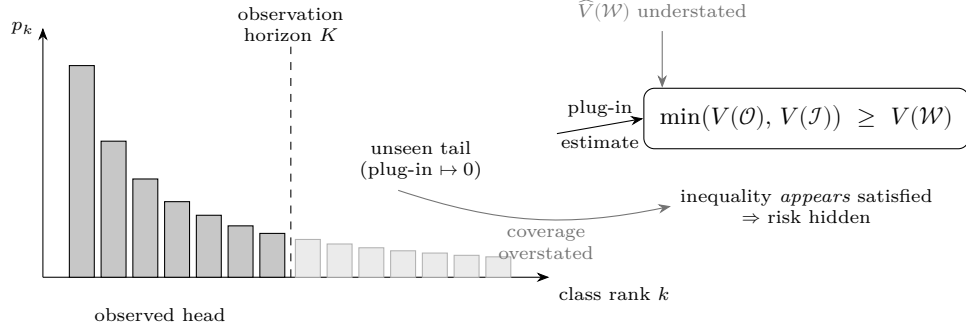


Figure 1: **The requisite-variety gap.** The plug-in estimator sees only the observed head of the outcome-relevant disturbance-class distribution and assigns zero probability to the unseen tail (shaded). It therefore *understates*  $V(W)$  and *overstates* coverage at once, so Wang et al.’s Principle-2 inequality  $\min(V(\mathcal{O}), V(\mathcal{J})) \geq V(W)$  can read as satisfied precisely when it is not. The Good–Turing missing mass  $\widehat{M}_n = N_1/n$  (§3.2) is the size of the shaded region.

partition-refinement procedure with three admission gates handles the case where residual entropy indicates that the disturbance partition itself, not the policy over it, is at fault.

§6 is the empirical evaluation. In the absence of production traffic for the demonstrator, we build a semi-synthetic benchmark — a known Zipf–Mandelbrot disturbance distribution over 149 LLM-authored task classes, driving a real agent (`single_react`) on a real model (`gpt-4o-mini`) with real tools — so that every estimator can be checked against a quantity we actually know. It also documents a measurement inversion caused by a regulatory bug, which we retain as a worked example of §4.5.

§7 treats the residual case that no statistic computed on the agent’s own logs can resolve, and its architectural consequence.

## 2. Background and Position

### 2.1 What we take from Agent Cybernetics

We adopt Wang et al.’s six principles without amendment, and their notation where possible. Two do most of the work here.

**Principle 2 (Requisite Variety).**  $\min(V(\mathcal{O}), V(\mathcal{J})) \geq V(W)$ . Output variety is the set of distinguishable responses, decisions and action sequences the agent can produce; internal variety is the set of distinguishable internal states it can occupy; the binding constraint is the smaller.

**Principle 6 (Context Entropy Minimization).** The context window is a finite-capacity channel:  $H(\text{output}) \geq \max(V(W) - C_{\text{channel}}, 0)$ . Whatever portion of disturbance entropy exceeds channel capacity is necessarily reflected in the output. Two notational points, since both matter later. We write the disturbance entropy as  $V(W)$  where Wang et al. write  $H(E)$ , because we reserve  $E$  for the essential variables of §3.1. And we take the positive part explicitly: output entropy is non-negative regardless, so the bound binds only when  $V(W) > C_{\text{channel}}$ . Where capacity is ample the principle imposes no floor at all, and residual output entropy must then be attributed to the regulator rather than the channel — which is the distinction §5.1 measures and §7 shows cannot always be made from inside the loop.

We take both as given. Our claim is not that these are wrong but that they are, at present, unevaluable.

**Wang et al.’s reliability taxonomy.** Their first desideratum bundles four peer failures — hallucination, feedback blindness, cascading failure, and irreversibility — as compositional, not as a single ranked headline mode. Their domain failure modes are then mapped onto principles rather than onto the four: feedback blindness appears as an instance of P1 (FM-CG1 in code, FM-CLI1 as exit-code blindness), goal drift as

an instance of P3 (FM-CG2, FM-AR2), insufficient action variety as P2 (FM-CG3), and irreversibility blindness and citation cycling as instances of P5, second-order regulation (FM-GUI3, FM-AR3). Their own practitioner ordering (Appendix A.6) puts P1 first as a structural prerequisite, but ranks **P5 meta-cognitive monitoring as the highest-value, lowest-cost intervention** — ahead of variety amplification and ahead of any feedback-blindness-specific fix.

We locate our instruments against that ordering explicitly.  $\beta$ ,  $\eta$  and the Good–Turing estimators address **P2**;  $\Phi$  addresses **P1**;  $H(R \mid D)$  addresses an organisational property the six laws do not express (§2.2). Our contribution to **P5** is not a monitor but a constraint on all of them: §7 shows that a second-order monitor computed on the agent’s own telemetry is provably blind to an entire failure class, which qualifies the “lowest-cost” half of their ranking — the cheap version of P5 is the version that cannot work. We do not measure goal drift (P3), classify irreversibility, or detect action-history loops, and we mark these as gaps rather than claiming them.

## 2.2 What is missing from the six laws

Two omissions matter for what follows.

The **Conant–Ashby good regulator theorem** (1970) is not among the six. This is a substantive gap: requisite variety constrains the *size* of the regulator’s repertoire, but the good regulator theorem constrains its *organisation* — the entropy-minimising regulator is a deterministic function of the disturbance, and its optimal mapping induces a homomorphism from the regulated system onto the regulator. The first result yields a directly measurable diagnostic,  $H(R \mid D)$ , which we develop in §5.1. Its omission means the framework has no account of a regulator that possesses ample variety and deploys it inconsistently — an agent that has the right tool and does not reliably reach for it.

**Beer’s Viable System Model** is likewise absent; Principle 5 (Second-Order Regulation) is von Foerster’s, not Beer’s. The consequence appears in §7: the framework has no analogue of Beer’s audit channel, and therefore no account of the failure class that is invisible to an agent’s own telemetry.

## 2.3 Related work

The good regulator theorem has been sharpened considerably. Richens & Everitt (2024) show that agents achieving low regret across a family of interventions must have learned an approximate causal model of the data-generating process; Richens et al. (2025) extend this to multi-step goals; Virgo et al. (2025) reformulate the theorem for embodied agents under partial observability; Wentworth’s information-bottleneck variant establishes conditions under which an optimal regulator must construct a model internally rather than merely behave as one. We take these as settled and do not restate them. Our concern is orthogonal: not *what an agent must have learned*, but *what an operator can measure about a deployed system*.

Endogenous distribution shift — the environment changing in response to the deployed policy — is **performative prediction** (Perdomo et al., 2020), with the distinction between performative *stability* and performative *optimality* directly relevant to §4.4. The exploration/safety tension is **dual control** (Feldbaum, 1960). The impossibility of detecting latent confounding from a single observational distribution, and the recovery of detectability under multiple environments (Karlsson & Krijthe, 2022), is the formal ancestor of §7. We claim no novelty in any of these; we claim that none of them has been turned into an operator-facing instrument for agent deployments, and that Agent Cybernetics — which calls for exactly such instruments — supplies the right frame in which to build one.

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## 3. Estimating $V(\mathcal{W})$ in an Open World

### 3.1 The partition

Ashby’s own move makes this tractable: do not enumerate raw states, enumerate **equivalence classes under outcome-relevance**. Fix the essential variables  $E$  — the quantities that must remain within bounds

(task success, cost ceiling, latency SLO, policy invariants). Two situations belong to the same *disturbance class*  $d \in \mathcal{D}$  iff they demand the same regulatory response to keep  $E$  in bounds.

This is not a convenience. It is the definition that makes the requisite-variety inequality evaluable at all, and it has a consequence we return to in §5.3: *any residual variation in response within a class is, by construction, either regulator inconsistency or evidence that the partition is wrong.*

In practice one rarely starts from the outcome-relevant partition. The initial ledger is typically a coarse, often *semantic* grouping of situations (a topic taxonomy, an intent schema), and a semantic topic is not automatically a response-equivalence class. Bringing the two into alignment — merging topics that demand the same response and splitting those that do not — is exactly the job of the partition-refinement procedure of §5.3; until it has run,  $\beta$  and  $H(R \mid D)$  measured on the coarse partition are read as provisional.

$V(\mathcal{W})$  is then the entropy of the class distribution, not the class count.

### 3.2 The bias of the plug-in estimator

Let  $n$  events yield  $K$  observed classes with counts  $\{c_1, \dots, c_K\}$ . The maximum-likelihood (plug-in) estimate

$$\hat{H}_{\text{ML}} = - \sum_i \frac{c_i}{n} \log_2 \frac{c_i}{n}$$

is biased downward, by approximately  $(K-1)/(2n \ln 2)$  (Miller–Madow), and the bias grows with the number of classes relative to sample size. But the more consequential error is not in the entropy estimate; it is in **coverage**.

Coverage computed over observed classes is, by construction, blind to the classes not yet observed. The Good–Turing estimate of the missing mass,

$$\hat{M}_n = \frac{N_1(n)}{n}$$

where  $N_1(n)$  is the number of classes observed exactly once, gives the probability that the *next* event falls in a class never before seen. Since an unobserved class is by definition unmapped,  $\hat{M}_n$  is a direct lower bound on the rate at which the deployed agent will face situations for which its ledger prescribes no response.

**Both biases run the same way.** The operator using plug-in estimation believes environmental variety is lower than it is, and coverage higher than it is. The requisite-variety inequality appears satisfied when it is not. We quantify this on real telemetry in §6.

### 3.3 Estimators

We use three, and report all three, because their disagreement is itself diagnostic:

- **Plug-in.** A lower bound. Reported only for comparison.
- **Good–Turing** (Simple Good–Turing, Gale & Sampson 1995): log-linear smoothing of the frequency-of-frequencies, redistributing mass  $\hat{M}_n$  to the unseen.
- **Chao–Shen** (2003): a Horvitz–Thompson estimator with Good–Turing coverage correction; better-behaved for small  $n$  with heavy tails.

Wide divergence between plug-in and Good–Turing is a signal that the class space is far from enumerated, independent of what the Heaps exponent says.

### 3.4 Discovery rate

Track the accumulation of distinct classes against events,  $D_n = Kn^\beta$  (Heaps’ law), and estimate  $\beta$  by log-linear regression on the discovery curve.

| $\beta$         | Regime    | Consequence                                             |
|-----------------|-----------|---------------------------------------------------------|
| $\rightarrow 0$ | saturated | class space effectively enumerated;<br>ledger converges |
| $\approx 0.5$   | partial   | typical of natural event streams                        |
| $\rightarrow 1$ | open      | classes arriving at a rate that no<br>ledger tracks     |

$\beta$  and the Zipf tail index  $\alpha$  of the class-frequency distribution are related ( $\beta \approx 1/\alpha$  for  $\alpha > 1$ ). §4 turns this into a decision procedure.

## 4. Enumeration Elasticity and the Futility Criterion

### 4.1 The question Wang et al. leave open

*How should an agent respond when its environment exceeds its representational capacity?* The framework offers two levers — amplify the regulator’s variety, or attenuate the environment’s. It does not say when to reach for which. We show the choice is decidable from the measured tail index, and that in one regime the amplification lever does not work at any feasible scale.

### 4.2 Statement

Consider an agent whose regulator is a mapping  $\rho : \mathcal{D} \rightarrow \mathcal{R}$  over an enumerated ledger of disturbance classes. *Amplification* means extending the ledger: mapping more classes to (possibly new) responses. Assume class frequencies follow a Zipf–Mandelbrot law with tail exponent  $\alpha > 1$ , related to the Heaps exponent by  $\beta = 1/\alpha$ .

Mapping a class is not free. Gate 2 requires a human or LLM judge to name it; Gate 3 requires a controlled experiment (§5.3). Write  $c$  for the cost of mapping one class and  $V$  for the value of correctly regulating one unit of probability mass. A rational operator maps the  $K$  most frequent classes, where  $K$  is chosen so that the marginal class still pays for itself.

**Theorem (Enumeration Elasticity).** *Let  $M^*$  be the residual — the probability mass left unmapped at the economically optimal ledger size  $K^*$ . Then the elasticity of the residual with respect to enumeration investment is*

$$\eta \equiv -\frac{\partial \log M^*}{\partial \log(V/c)} = \frac{\alpha - 1}{\alpha} = \boxed{1 - \beta}$$

*Proof in Appendix A. The result is exact, not asymptotic in  $V/c$ .*

### 4.3 Reading it

The elasticity is stated **entirely in terms of the Heaps exponent** — the quantity §3.4 already measures directly from the discovery curve, requiring no separate estimation of the tail index and no assumption about where the head of the distribution sits.

It says: *a  $\lambda$ -fold increase in enumeration investment reduces the unhandled fraction of traffic by a factor of  $\lambda^{1-\beta}$ .* Equivalently, the investment multiple required to **halve** the residual is  $2^{1/(1-\beta)}$ :

| $\beta$ | $\eta = 1 - \beta$ | investment multiple to halve residual |
|---------|--------------------|---------------------------------------|
| 0.3     | 0.70               | 2.7×                                  |
| 0.5     | 0.50               | 4×                                    |
| 0.7     | 0.30               | 10×                                   |
| 0.8     | 0.20               | 32×                                   |

| $\beta$ | $\eta = 1 - \beta$ | investment multiple to halve residual |
|---------|--------------------|---------------------------------------|
| 0.9     | 0.10               | 1,024×                                |
| 0.95    | 0.05               | $\sim 10^6 \times$                    |

**As  $\beta \rightarrow 1$ ,  $\eta \rightarrow 0$ : investment stops buying coverage.** Not asymptotically, not eventually — the elasticity is the exponent, and it goes to zero. At  $\beta = 0.9$ , an organisation that increases its class-mapping effort by three orders of magnitude halves the fraction of traffic it cannot handle. No engineering organisation will do that, and if it did it would still be unhandling half of what it was unhandling before.

This is not an impossibility result and we do not overclaim it as one. For a fixed stationary distribution the missing mass tends to zero almost surely: enumerate long enough and you will eventually see everything. The claim is economic, and sharper for being so. **Enumeration remains possible in the limit and stops being purchasable in the deployment.**

One simplification is optimistic in the same direction. The theorem assumes a *constant* per-class mapping cost  $c$ ; but the classes still unmapped at large  $K$  are precisely the low-frequency, ambiguous, noisy ones that are hardest to isolate and adjudicate, so  $c$  almost certainly rises with  $K$ . Appendix A.6 relaxes the constant-cost assumption, and the effect runs one way only: super-linear mapping cost steepens the residual curve, so the constant- $c$  elasticity above is a *best case* and the futility of pure enumeration in the  $\beta \rightarrow 1$  regime is, if anything, understated here.

#### 4.4 The decision rule

The two levers of Principle 2 are no longer a matter of taste. Measure  $\beta$ ; the elasticity follows, and with it the lever (Figure 2):

- $\beta \lesssim 0.5$  ( $\eta \gtrsim 0.5$ ). Amplification works. Extend the ledger, map the classes; each addition returns real coverage, and the residual falls roughly as the square root of investment. This is the regime the framework implicitly assumes throughout.
- $\beta \rightarrow 1$  ( $\eta \rightarrow 0$ ). The marginal class carries vanishing mass and the return on enumeration collapses. Requisite variety **cannot be satisfied by amplification at any ledger an operator will rationally build**. The deficit must be met on the other side of the inequality: by attenuation that provably shrinks  $V(\mathcal{W})$  (schemas, scope reduction, input constraint), or by a *class-independent* response whose coverage does not depend on having anticipated the class — graceful degradation, human escalation, delegation to a higher level of recursion.

That last clause is the operational content of Wang et al.’s open question. *How should an agent respond when its environment exceeds its representational capacity?* When  $\eta \approx 0$ , the answer is: **not by trying to represent more of it**. The failure is not that some particular class was missed; it is that the class space has a tail index under which no anticipation strategy is worth funding. The correct response is architectural — a fallback whose coverage is independent of the partition — and the criterion tells you when you have crossed into needing one.

#### 4.5 A caution: the criterion can be gamed by the system it measures

The disturbance distribution an agent faces is not exogenous. An agent that attenuates aggressively — narrowing its scope, refusing ambiguous inputs, constraining what it will accept — *shrinks the variety it sees*. Its measured  $\beta$  falls, its elasticity rises, its coverage improves, and its ledger converges. It will appear, on every instrument in this paper, to be getting better.

This is **performativity** (Perdomo et al., 2020), and it means a converged ledger is ambiguous between two opposite conditions: the agent has mastered its environment, or the agent has shrunk it. These are the *performatively optimal* and *performatively stable* solutions respectively, and the distinction is exactly Perdomo et al.’s.

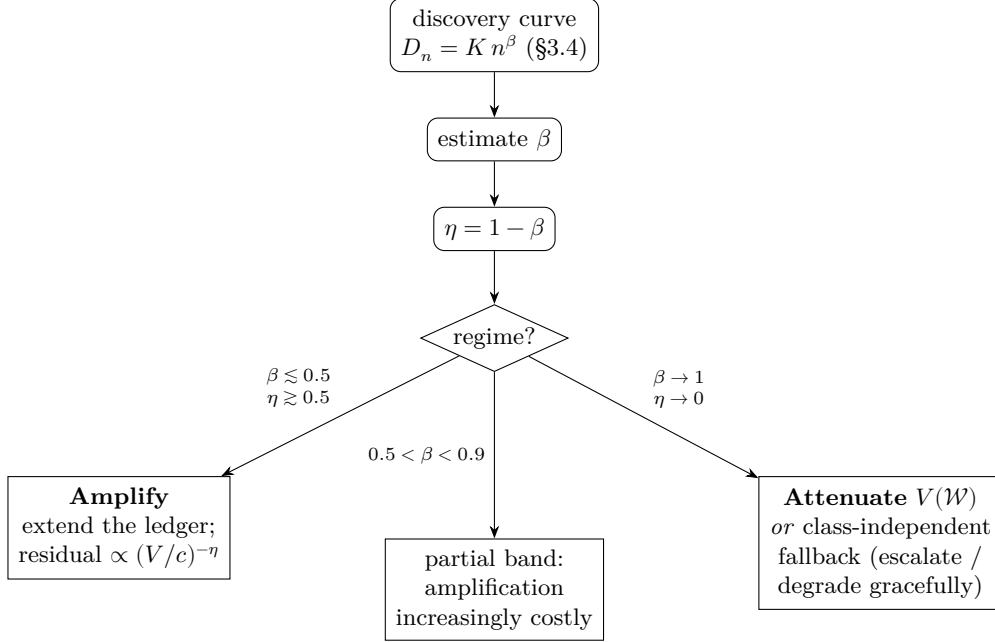


Figure 2: **The enumeration-futility decision rule (§4.4).** The Heaps exponent  $\beta$ , read directly off the discovery curve, fixes the elasticity  $\eta = 1 - \beta$  and thereby which lever of Principle 2 to pull. The investment multiple required to halve the unhandled residual is  $2^{1/(1-\beta)}$  — modest for  $\beta \lesssim 0.5$ , prohibitive as  $\beta \rightarrow 1$ , where amplification is abandoned for attenuation or a class-independent fallback.

The hazard is most acute for our own criterion. §4.4 recommends attenuation when  $\beta \rightarrow 1$  — and attenuation lowers measured  $\beta$ . **The recommended treatment improves the diagnostic whether or not it improves the system.** An operator following this paper’s advice without the mitigation below will observe their numbers getting better and will have no way to tell whether anything did.

We do not resolve this. The mitigation is that  $\beta$  and coverage must be reported against a **fixed held-out disturbance stream** — a canary set sampled before the policy was deployed and never filtered by it — not against live traffic the policy has already shaped. Any deployment of the variety ledger without such a canary is measuring its own reflection. We regard this as the most important practical caveat in the paper.

## 5. Three Further Instruments

### 5.1 $H(R | D)$ — the determinism instrument

The Conant–Ashby lemma states that the entropy-minimising regulator’s response is a deterministic function of the disturbance: any residual  $H(R | D) > 0$  is unnecessary noise, and all responses assigned with non-zero probability to the same disturbance yield the same outcome anyway, so there is no reason to have more than one.

Because disturbance classes are *defined* by response-equivalence (§3.1),  $H(R | D)$  measured on agent logs is a direct test of this condition, and it is a diagnostic the six laws cannot express. We call the failure it detects **dithering**: the same disturbance class mapped to different responses across occurrences.

Dithering is expensive in a specific way. The same input produces different downstream cost, latency and success purely as a function of which branch was taken, not of anything different about the situation. It is variance that buys nothing — unlike exploration, which trades consistency for information.

Two causes, requiring opposite fixes:



1. **The policy is loose.** Soft probabilistic dispatch where hard routing was warranted. Fix: tighten  $\rho$ .
2. **The partition is too coarse.** The class conflates situations that demand different responses. Fix: split the class (§5.3).

$H(R \mid D)$  detects the condition. It cannot distinguish the causes. §5.3 is the procedure that does.

## 5.2 Feedback blindness, made falsifiable

Wang et al. name *feedback blindness* among four compositional reliability failures; they diagnose it in domain-specific form (e.g. FM-CG1, FM-CL11) but do not measure it. The failure is:  $a_{t+1}$  conditioned on  $c_t$  but not causally on  $y_t$  — the agent appends the tool result to context and regenerates the prior plan.

The natural observational statistic,  $I(a_{t+1}; y_t \mid c_t) \approx 0$ , is a trap. It is confounded: agents that receive failure signals are, by selection, in harder states, and  $c_t$  is high-dimensional enough that conditioning on it properly is not achievable from logs.

The test must be interventional, and it is cheap. **Ablate the observation and resample the action.** For a matched set of steps, replace  $y_t$  with a neutral placeholder, re-run the policy, and measure the divergence between the action distributions:

$$\Phi = \mathbb{E} [D_{\text{KL}}(\pi(\cdot \mid c_t, y_t) \parallel \pi(\cdot \mid c_t, \tilde{y}_t))]$$

$\Phi \approx 0$  is feedback blindness, established by intervention rather than inferred from correlation. It is a rung-2 quantity in Pearl’s sense, it is computable against any agent one can re-run, and it converts a named but unmeasured reliability failure from a diagnosis into a number with a threshold. We propose it as a standard reliability metric.

## 5.3 Partition refinement and its three gates

When  $H(R \mid D)$  is high for a class, the mechanical move is decision-tree induction: search exogenous features for a binary split that reduces conditional entropy. Information gain, ID3/C4.5, nothing exotic.

The difficulty is admission. A split search will find structure that is statistically real and semantically vacuous, and a framework that promotes such splits inflates its own  $\beta$  by fitting noise as structure — manufacturing the open-world pathology it exists to detect. We impose three gates (Figure 3).

**Gate 1 — Statistical.** Minimum leaf size; G-test of independence (preferred over Pearson  $\chi^2$  for the sparse contingency tables agent logs invariably produce); MDL penalty requiring information gain to exceed the description-length cost of the added ledger complexity.

**Gate 2 — Policy improvement, not outcome differential.** *This gate corrects a wrong instinct, and the correction is the substantive methodological claim of this section.*

The intuitive screen — do the two branches have different success rates? — produces false negatives on exactly the splits that matter most. Consider a class whose coarse default response is already correct for the majority branch and wrong only for a minority branch. The minority is badly served; the aggregate branch-versus-branch success gap is diluted; the marginal test rejects a split that is necessary. We observed precisely this in development: a class with a genuine latent bifurcation was rejected on a 3.7pp marginal delta because its default response was already correct for 85% of the class.

The correct criterion is **regret**, evaluated per branch:

$$\text{Regret}(b) = \max_{r \in \mathcal{R}} \text{succ}(b, r) - \text{succ}(b, r_{\text{default}})$$

*Is the current coarse mapping leaving value on the table for this sub-population?* If yes, the class conflates situations demanding different responses — Ashby’s definition of a partition that should be split. If no, the

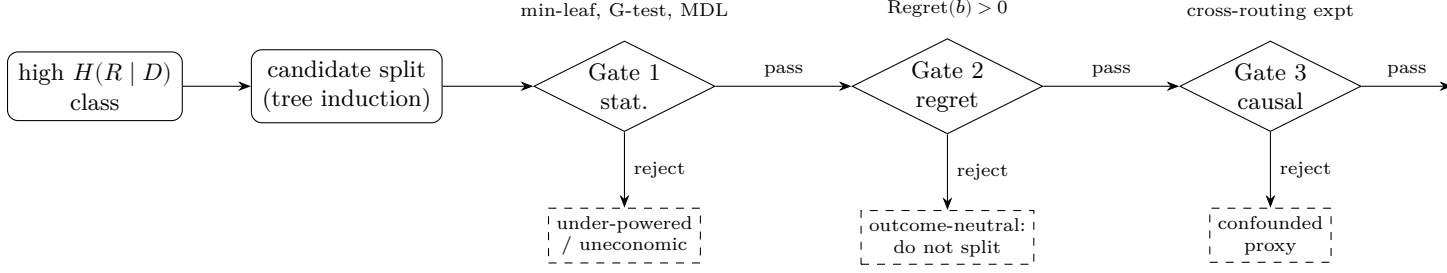


Figure 3: **The three admission gates for partition refinement (§5.3).** A high-dithering class yields a candidate split that must clear all three gates to enter the ledger. Gate 1 rejects statistically under-powered or MDL-uneconomic splits; Gate 2 rejects splits that are real but *outcome-neutral* (no per-branch regret, hence no policy improvement); Gate 3 rejects features that are confounded proxies, requiring an interventional cross-routing experiment. The stopping rule is Ashby’s: split only until it would change the required response.

coarse response is already optimal for both branches, the split would not change what the agent *does*, and by the stopping rule it must not be made however large the information gain.

This is the contextual-bandit view of the same object: the coarse ledger is a context-free policy, and the split is warranted iff conditioning on context yields a policy improvement.

**Gate 3 — Causal.** Regret measured on logs is a counterfactual claim — *this branch would have done better under a different response* — estimated from data generated under the incumbent policy, which is where confounding bites. The feature may be a proxy for something the router already reacts to. Promotion to the ledger requires a controlled cross-routing experiment, whose sample size we compute from the observed regret. Every ledger mapping carries a rung tag (observational / interventional / counterfactual) recording what evidence underwrites it.

**The stopping rule** is Ashby’s own and it is the clean part: disturbance classes are *defined* as situations requiring the same response, so split until further splitting would not change the required response, and stop. Not at some arbitrary granularity.

#### 5.4 Susceptibility $\Sigma$ — the adversarial sibling of $\Phi$

§5.2’s  $\Phi$  asks how much the action distribution moves when the observation  $y_t$  is *ablated*. The dual question is what happens when  $y_t$  is *authored by an adversary*. Indirect prompt injection is exactly this: a data-input tool returns content that contains instructions, and the disturbance enters through the **observation channel**, not the input channel. It is the adversarial instance of the object  $\Phi$  intervenes on, and it inherits  $\Phi$ ’s central lesson. If a policy conditions on whether a result is *usable* rather than whether its content is *trustworthy* (the §6.7 finding), then a fluent injected instruction is maximally usable and the policy is structurally exposed.

We define **susceptibility**

$$\Sigma = P(\text{the agent complies with an instruction injected into } y_t)$$

over full trajectories, measured against a matched **clean control** — the identical task with an un-poisoned observation — so that  $\Sigma_{\text{clean}} \approx 0$  is a required sanity check rather than an assumption.  $\Sigma$  is directed where  $\Phi$  is symmetric:  $\Phi$  measures sensitivity to *any* change in  $y_t$ ;  $\Sigma$  measures capture by a *specific adversarial* change. The §6.7 prediction is then a falsifiable relation between the two — a policy that is feedback-*insensitive* to result content (low  $\Phi$  under corruption) should be *more* susceptible (high  $\Sigma$ ), because it will not distinguish an injected instruction from legitimate content.

$\Sigma$  is a variety statement. The injection is an adversarial disturbance class; the safety-relevant response repertoire is {resisted, partial, complied} with **complied** a *forbidden* response; and MI(attack class; response) is the variety the adversary succeeds in pushing across the observation boundary. A robust regulator drives

$H(\text{safety} \mid \text{attack})$  and this mutual information toward zero — the Conant–Ashby condition, now under attack. Because the attack surface is itself open-world (§4), the enumeration-futility criterion applies to the *defense*: a denylist enumerated over attack instances has elasticity  $\eta = 1 - \beta$  against an adversary who can generate unbounded phrasings of a fixed payload, which is the formal case for a class-independent guardrail over signature matching.

## 6. Empirical Evaluation

### 6.0 What this benchmark is, and what it is not

The paper turns on this section, and the real constraint is that the demonstrator agent has no production user traffic. We therefore cannot claim to have measured *the* Heaps exponent of real agent deployments. Rather than substitute a fully synthetic log — which would only show that our estimators recover structure we planted — we constructed a **semi-synthetic** benchmark in which the *disturbance-arrival process is controlled and known* while the *response, outcome and cost data are genuinely produced by a model executing*:

1. **Taxonomy.** An LLM authored 149 human-readable task categories across 15 domains (current events, math/stats, coding, travel, creative writing, ...), each with three concrete example prompts. These are the disturbance classes  $\mathcal{D}$  — but as a *semantic* partition, not yet the outcome-relevant one of §3.1. Treating each topic as a disturbance class is a deliberate coarse starting point, so the E2/E3 readings below are of a semantic ledger *before* refinement, and the  $\beta$  and  $H(R \mid D)$  they report partly reflect the authoring model’s ontology; §6.4 runs the partition-refinement gates against exactly these classes to test how much of that structure survives an outcome-relevance criterion.
2. **Known class distribution.** Each class was assigned a Zipf–Mandelbrot frequency  $p_k \propto (k + q)^{-\alpha}$  with  $\alpha = 1.5$ ,  $q = 2.7$  — the exact generalisation Appendix A.6 proves is elasticity-invariant. This fixes a *known* true entropy  $H(\mathcal{D}) = 5.373$  bits and a known unseen-mass function at every  $n$ .
3. **Real execution.** We sampled a stream of 2,000 events from that distribution and fired each prompt at the live backend: the `single_react` agent (`gpt-4o-mini`, web-search + Python-REPL tools). The response variable is *what the agent actually did* — which tool(s) fired, or a structural classification of a no-tool answer — and cost/latency are the real figures.

So E1/E2 test the estimators against a ground truth we possess; E3/E5 test whether real model behaviour, driven by a known input distribution, contains the regulatory structure the instruments are designed to find. The benchmark is a controlled analogue of a deployment, not a deployment. The reference implementation and data-generation scripts are open-sourced (`experiments/paper/`).

The following results are the **corrected** run ( $n = 2,000$ ). A prior run of the identical benchmark produced systematically different numbers because of a regulatory defect in the agent itself; §6.6 reports that incident, because it is a live instance of §4.5 rather than a footnote.

#### 6.1 E1 — Estimator bias

At  $n = 2,000$  the agent had encountered 138 of 149 classes (92.6% coverage). Entropy estimates against the known  $H = 5.373$  bits:

| Estimator   | $\hat{H}$ (bits) | bias vs. truth |
|-------------|------------------|----------------|
| Plug-in     | 5.336            | <b>−0.037</b>  |
| Good–Turing | 6.315            | <b>+0.942</b>  |
| Chao–Shen   | 5.421            | <b>+0.048</b>  |

At near-saturation the plug-in estimate is *nearly unbiased* and Good–Turing badly *over-corrects* — the reverse of the textbook warning. This is not a refutation of §3; it is the expected behaviour of these estimators when  $n$  is large relative to the class space, and it is why we insist all three be reported: their divergence is the

signal. The predicted plug-in pathology reappears exactly where §3.2 says it should — in the undersampled regime. Splitting the stream at  $n = 1,000$  (train) /  $n = 1,000$  (held-out):

| quantity                                                           | value        |
|--------------------------------------------------------------------|--------------|
| true $p(\text{unseen})$ after 1,000 events                         | 3.34%        |
| Good–Turing estimate $\hat{M}_n$                                   | 4.10%        |
| <b>actual</b> realised new-class fraction in the untouched holdout | 3.80%        |
| plug-in implied $p(\text{unseen})$                                 | <b>0.00%</b> |

Good–Turing tracks the true and the empirically realised unseen mass to within a point; the plug-in estimator asserts, with confidence, that there is nothing left to see. This is §3.2’s “both biases run the same way” claim, demonstrated against a known answer: in the regime where it matters — before the class space is exhausted — plug-in coverage is not conservative-but-crude, it is categorically wrong in the direction that hides risk.

## 6.2 E2 — Tail index and elasticity

Fitting Heaps’ law to the discovery curve gives  $\hat{\beta} = 0.458$  on the full  $n = 2,000$  curve and  $\hat{\beta} = 0.518$  on the first half, against the *generative* value  $1/\alpha = 0.667$ .

The fitted  $\beta$  is below the theoretical value, and the gap widens as  $n$  grows — this is the characteristic signature of a **finite** vocabulary, not estimator failure. The theory’s  $\beta = 1/\alpha$  assumes an unbounded class space; our taxonomy has exactly 149 classes, 138 already seen, so the discovery curve is visibly bending toward its asymptote and the local slope is falling. This is itself the point of §3.4 and §4.3: **the measured  $\beta$  tells you the regime you are in, and here it correctly reports a large-but-closed world approaching saturation** — precisely the distinction (finite vs. genuinely open support) that Appendix A.6 flags as undecidable from a converged ledger alone and that the Good–Turing mass in E1 is needed to disambiguate.

Applying §4: at  $\hat{\beta} = 0.458$ ,  $\eta = 1 - \beta = 0.542$ , so halving the residual costs a  $2^{1/\eta} \approx 3.6\times$  increase in enumeration effort — squarely in the “amplification works” band ( $\beta \lesssim 0.5$ ). The generative  $\beta = 0.667$  ( $\eta = 0.33$ ,  $\approx 10\times$  to halve) sits in the harder “partial” band. That the *measured* and *generative* regimes differ is the empirical content: an operator reading  $\hat{\beta}$  off a saturating finite stream would over-estimate the return on enumeration, which is why §4.4 conditions its decision rule on a canary (§6.5), not on live saturation.

## 6.3 E3 — Dithering

Overall  $H(R | D) = 0.200$  bits: most classes are regulated deterministically (the majority response is “answer directly, no tool”). Ten classes carry non-trivial residual entropy, and they are exactly the classes where the correct regulatory response is underdetermined by the coarse policy:

| class                                   | $H(R   D)$ | $n$ | response split               |
|-----------------------------------------|------------|-----|------------------------------|
| business_finance_calculate_roi          | 0.985      | 14  | 6 direct / 8 Python-REPL     |
| math_stats_computation_basic_arithmetic | 0.973      | 297 | 177 direct / 120 Python-REPL |
| math_stats_computation_probability      | 0.971      | 35  | 21 direct / 14 Python-REPL   |
| comparisons_tradeoffs_product_features  | 0.946      | 11  | 4 direct / 7 web-search      |

| class                                         | $H(R   D)$ | $n$ | response split                   |
|-----------------------------------------------|------------|-----|----------------------------------|
| coding_data_<br>transform_transform_<br>value | 0.943      | 25  | 16 direct / 9<br>unexecuted-code |

This is dithering in Conant–Ashby’s exact sense: the same disturbance class mapped to different responses across occurrences, at cost (a Python-REPL branch spends tokens and latency a direct answer does not) with no corresponding difference in the situation. It is concentrated on computation-shaped tasks, where the agent is inconsistent about whether to *actually run* a calculation or answer from the model’s arithmetic — a substantive and operator-relevant defect that the instrument localises without being told where to look.

**Variety balance.**  $V(\mathcal{D}) \approx 6.3$  bits (Good–Turing) against  $V(\mathcal{R}) = 2.32$  bits over five effective responses. The requisite-variety inequality  $V(\mathcal{R}) \geq V(\mathcal{D})$  is violated by roughly four bits — the agent’s response repertoire is far narrower than the variety of situations it faces, which is the expected condition for a single small-model agent and is now a measured quantity rather than an assertion.

#### 6.4 E5 — Partition refinement and its gates

The refinement engine ran against the ten high- $H(R | D)$  classes with the exogenous features the benchmark logs (prompt variant, prompt length, domain). Nine candidate splits were rejected at **Gate 1** (minimum leaf size) and one at **Gate 3** (no policy improvement). None was promoted.

The binding constraint here is operational and worth stating precisely: the median class received 4 events, and only 8 of 138 classes cleared 60 events, so `min_leaf = 30` correctly rejects almost everything as under-powered *before feature choice is even in play*. This is the intended behaviour of Gate 1 — at this volume the correct answer is “insufficient evidence to split,” and the engine returns it — but it means E5 cannot, at this  $n$ , exhibit the more interesting Gate-2 vs. Gate-3 contrast the design is built for. That contrast additionally requires a graded outcome signal: the benchmark’s `task_success` (did the run complete with non-empty output) was 1.0 for all 2,000 events, so the outcome-differential and regret computations were structurally degenerate under it. We therefore added one.

**A graded outcome signal (LLM-as-judge).** We re-ran the identical 2,000-event stream capturing each agent’s *full* answer, and graded every (prompt, answer) pair with a **stronger** judge model (`gpt-4o`, deliberately above the `gpt-4o-mini` under test to limit self-preference bias) on a three-level rubric {incorrect = 0, partial = 0.5, correct = 1}. The resulting `answer_quality` is non-degenerate — {0 : 89, 0.5 : 68, 1 : 1843}, **mean 0.939** — and drops straight into the ledger’s `outcome_col`.

With it, two things change. First, the outcome channel is no longer flat: outcome entropy rises to  $H(E) = 0.474$  bits and **20 classes leak** ( $H(E | D) > 0.5$ ). The incident-attractors the graded signal exposes are *not* the arithmetic classes but the current-events and data-shaping ones, where answering from memory or narrating unexecuted code fails: `current_events...latest_headlines` (success 0.067,  $n = 15$ ), `coding_data_transform_format_data` (0.155,  $n = 42$ ), `current_events...political_updates` (0.318,  $n = 33$ ). This is the outcome-leakage map the paper’s essential-variable machinery is for, and it was invisible under `task_success`.

Second, the Gate-3 regret screen becomes non-vacuous: it now runs on *real* success rates. The strongest candidate — splitting `coding_data_transform_transform_value` — shows a substantial branch outcome gap (0.375 vs 1.00,  $p = 1.6 \times 10^{-4}$ ). And yet **every candidate still rejects**, for a reason that is now substantive rather than an artifact of a constant outcome: **per-class  $\times$  per-response power**. The dithering mid-tail classes are small ( $n = 14$ –35); the regret test needs  $\geq 15$  events for a *second* response within a branch to name a superior alternative, which they cannot supply, so `max_regret` is legitimately 0. A `min_leaf = 15` sensitivity pass reaches the identical verdict as `min_leaf = 30` ( $9 \times \text{REJECTED\_MIN\_LEAF}$ ,  $1 \times \text{REJECTED\_NO\_POLICY\_IMPROVEMENT}$ ); no proposal reaches `PENDING_SEMANTIC`, so the implemented LLM-driven Gate-2 naming step has nothing to name — itself the correct reading at this volume.

**Sensitivity (stratified booster pass).** To test whether those `min_leaf`-dominated rejections were power-limited rather than substantive, we topped the 17 highest- $H(R | D) / H(E | D)$  classes to  $n = 60$  via flagged booster events (+691 runs appended after the base stream; the 2,000-event Zipf sub-stream is unchanged for E1/E2). Classes clearing Gate 1 rose from 8 to 24; at `min_leaf = 15`, rejections shifted from  $9 \times \text{REJECTED\_MIN\_LEAF}$  to  $6 \times \text{REJECTED\_NO\_POLICY\_IMPROVEMENT}$ ,  $4 \times \text{REJECTED\_MIN\_LEAF}$ , and  $2 \times \text{REJECTED\_NOT\_SIGNIFICANT}$  — confirming that Gate-3 regret screening activates once per-class power clears. Still no proposal reaches `PENDING_SEMANTIC`: `max_regret` remains 0 throughout, because the coarse default response is already optimal on both branches of every surviving candidate. That is Ashby’s stopping rule working as designed, not a failure of the machinery; it also shows the binding constraint at canonical volume was evidence, not a broken gate.

The single most telling result ties E3 and E5 together. The largest and highest-dithering class, `math_stats_computation_basic_arithmetic` ( $n = 297$ ,  $H(R | D) = 0.973$ ), has a graded outcome of **1.0 on all 297 events**: the agent dithers over *method* — run Python or answer directly — but both methods get the arithmetic right, so its dithering is outcome-neutral and a split there correctly yields zero regret. **Response dithering is not the same as outcome leakage**, and the graded ledger distinguishes them exactly as Ashby’s stopping rule requires: do not split where it would not change what you *do*. The classes that warrant attention are the leaking ones, not the dithering ones — and only a real outcome signal can tell them apart.

The judge is itself a model, and its adequacy — especially on the current-events classes it scores lowest, where it too lacks fresh knowledge — is exactly the open question §8 flags. The graded outcome is strong enough to render E5’s gates operational; it should not be read as calibrated correctness.

## 6.5 E6 — Canary

The corrected 2,000-event stream is frozen as `canary_v1.jsonl` and versioned. Per §4.5 this is the policy-independent reference against which any future  $\beta$  or  $H(R | D)$  must be measured: without it, a later drop in  $\beta$  cannot be attributed between “the world got simpler” and “the policy started refusing or narrowing the traffic it sees.” §6.6 is the reason we treat this as structural rather than ceremonial.

## 6.6 A measurement inversion caused by the regulator — a worked §4.5 incident

The first run of this exact benchmark reported  $H(R | D) = 0.042$  bits overall and **zero actual tool calls across all 2,000 events**; the response variable collapsed almost entirely onto “answered directly” plus a residue of *narrated* tool intent (“I’ll search...”, unexecuted `python` blocks) that never resolved into a tool invocation. Read naïvely, that log says the agent is an extremely *deterministic* regulator — low dithering, a nearly one-to-one  $\mathcal{D} \rightarrow \mathcal{R}$  map.

It was an artifact. The agent’s model factory returned a *callable* (to support per-request model selection), and the agent framework binds tool schemas only to *static* models — a callable model is passed through with no tools attached. The regulator was not choosing not to act; it had been silently stripped of its entire action repertoire and could not act.  $V(\mathcal{R})$  was effectively collapsed to a single response, and every downstream statistic reported a system that looked cleanly regulated precisely *because* it had no variety to deploy.

After the defect was fixed, the identical stream — same taxonomy, same seed, same event order — produced 619 real tool invocations (291 web-search, 185 Python-REPL),  $V(\mathcal{R}) = 2.32$  bits, and  $H(R | D)$  rose from 0.042 to 0.200 bits, localising onto the computation classes of §6.3.

We name this failure **repertoire collapse**, because it has no slot in Wang et al.’s reliability taxonomy and we think it needs one. Their four Desideratum-1 failures concern the *content* of a response (hallucination), its *conditioning* on observations (feedback blindness), its *propagation* along a trajectory (cascading failure), and the *type* of action taken (irreversibility). What we observed is none of these. The repertoire itself was silently emptied —  $V(\mathcal{R})$  collapsed to a single response — while the agent went on narrating actions it no longer had the capacity to take, which is the surface signature to look for: tool intent that never resolves into a tool call. The diagnostic property that makes it dangerous is that it does not degrade the in-band metrics, it *improves* them. Detecting it means checking that the response variety the ledger declares is the response

variety the runtime actually holds — a property of the deployment, not of the trajectory, and therefore not recoverable from the trajectory at all (§7).

The methodological content is the whole point of §4.5, observed rather than argued:

- **A converged / low-entropy ledger is ambiguous between “well-regulated” and “incapable of varying.”** Here the *lowest* dithering reading corresponded to the *most broken* regulator. An operator watching  $H(R | D)$  fall would have recorded an improvement.
- **The estimators (E1/E2) were unaffected** — they measure the disturbance distribution, which is the controlled part — while the regulatory instruments (E3) inverted completely. That dissociation is exactly why the paper separates  $V(W)$  estimation from  $H(R | D)$ : they fail independently.
- **This is why measurement must be against a frozen canary and re-run after any policy change** (§4.5, §6.5). The corrected numbers are only interpretable *because* the disturbance stream was held fixed across the fix; had we also regenerated the stream, we could not have attributed the change to the regulator rather than the world.

We report the incident in full rather than quietly publishing the corrected table. A framework whose thesis is that agent regulation is systematically mis-measured should hold itself to the same standard, and the cheapest way to earn that claim is to have caught it happening to us.

### 6.7 E4 — Feedback blindness, measured

We built the interventional harness of §5.2 and ran it against the compiled `single_react` policy in-process. **Phase 1** ran the agent on 25 tool-forcing prompts (current-events, math/stats, comparisons) and captured, for each, the message prefix up to and including the *first* tool result — the context  $c_t$  plus the real observation  $y_t$  — together with the action the agent actually took next; all 25 fired a tool. **Phase 2** reconstructed the exact bound policy  $\pi$  (same system prompt, same tools, `bind_tools`) and resampled the next action  $K = 12$  times at temperature 0.8 under three conditions: the real  $y_t$ ; a neutral ablation  $\tilde{y}_t$  (“[no result returned]”); and a corrupted/contradictory result. Each sampled action was discretised to `{web_search, python_repl, final_answer}`. A **noise floor** — the divergence attributable purely to temperature sampling at finite  $K$  — was estimated by splitting the real-condition samples in half.

| quantity                                      | value                     |
|-----------------------------------------------|---------------------------|
| $\Phi$ (Jensen–Shannon, real vs. neutral)     | <b>0.553 bits</b>         |
| $\Phi$ (Jensen–Shannon, real vs. corrupt)     | 0.553 bits                |
| resampling noise floor (JS)                   | 0.006 bits                |
| signal-to-noise                               | $\approx 90\times$        |
| $\Phi$ (KL, Laplace-smoothed — the §5.2 term) | 3.04 bits                 |
| total variation                               | 0.66                      |
| decision points moving above noise            | 18 / 25                   |
| <b>verdict</b>                                | <b>feedback-sensitive</b> |

$\Phi$  sits roughly ninety times above the sampling-noise floor: **single\_react is decisively not feedback-blind** — the opposite of what §6.6’s broken regulator could have been mistaken for. The canonical decision point is unambiguous: for “*What is the most recent news about OpenAI’s product roadmap?*”, given a real search result the policy finalises ( $p(\text{final\_answer}) = 0.98$ ); ablate or corrupt that result and it re-searches ( $p(\text{web\_search}) = 0.96$ ). The next action is contingent on the observation.

Two nuances, both reported rather than smoothed over. (i) **The neutral and corrupt ablations move the action distribution by an identical 0.553 bits.** The policy conditions on whether a result is *usable*, not on whether its content is *correct*: an empty placeholder and a plausibly-wrong result both trigger the same “do not finalise, search again” reaction. A content-verification probe — does the agent detect that a returned result contradicts the query? — is a distinct and stronger test that this instrument does not yet isolate, and the equality is the finding that motivates it. (ii) 7 of 25 points show  $\Phi \approx 0$ : cases where the

agent re-searches regardless of the result (its first result was insufficient), so there is no residual sensitivity left to detect at that step.

**Caveats, baked into the results file.** Estimating an action *distribution* requires temperature  $> 0$ , whereas production `single_react` runs at temperature 0 —  $\Phi$  is measured on a deliberately stochastic variant of the policy, and establishes that it is feedback-*sensitive*, not that it is feedback-*correct*. The action is discretised at tool-choice granularity, not over answer content. Finite- $K$  KL is biased upward, so we lead with the bounded Jensen–Shannon divergence and always report  $\Phi$  against the measured noise floor rather than against zero. The natural next instrument extends  $\Phi$  from tool-choice granularity to a divergence over the *continuous generated text* — or over its embedded semantic intent, via an embedding distance between resampled answers — and would test feedback-*correctness* rather than merely feedback-*sensitivity*; the identical neutral-vs-corrupt divergence reported above is precisely the gap such a content-level  $\Phi$  would close, and we leave it to future work.

## 6.8 Threats to validity

This is a semi-synthetic benchmark on a single small model at  $n = 2,000$ ; none of E1–E5 should be read as a measurement of production agent traffic, and the finite 149-class vocabulary means E2’s regime finding is about *this* closed world. The outcome half of E5 is now exercised via a graded signal (§6.4), but that signal is an `gpt-4o` LLM-judge without a ground-truth key, and is least trustworthy on precisely the current-events classes it scores lowest, where the judge itself lacks fresh knowledge — it is adequate to make the gates operational, not to be read as calibrated correctness. Even with it, per-class power (median = 4 events/class,  $8/138$  classes  $\geq 60$ ) is the binding constraint on *which* splits are warranted, though conclusions about the *gates* themselves are sound. E4’s  $\Phi$  (§6.7) is measured at temperature 0.8 on 25 decision points at tool-choice granularity: it establishes feedback-*sensitivity*, not feedback-*correctness*, and the identical neutral/corrupt divergence shows content verification is unmeasured. The taxonomy was LLM-authored and may carry that model’s notion of “distinct task,” which is a partition assumption of the kind §8 warns is itself a model. The E7 poisoning results (§6.9) carry their own caveats: the safety signal is again a `gpt-4o` judge (though it agrees with the deterministic canary markers to within 3pp on the marker-detectable cases, which bounds its unreliability where it can be checked);  $\Sigma$  is measured at temperature 0.7, so it reflects a stochastic variant of the data-tool policy; the payloads are deliberately benign canaries, so  $\Sigma$  measures *susceptibility*, not end-to-end *exploitability*; the web vector is a fixture (no live web, no egress) whereas the file and RAG vectors are fully real; and the attack taxonomy is a small hand-authored plus LLM-expanded set, so the Enumeration-Elasticity  $\beta$  over our own instances saturates and the enumeration-futility argument rests on the *structural* unboundedness of the phrasing surface, not on a large measured  $\beta$ . What the benchmark does establish is that the instruments compute correctly against known ground truth, that real model behaviour under a known input distribution contains recoverable dithering *and outcome-leakage* structure the instruments localise unaided, that the feedback-blindness test discriminates a responsive policy from noise at  $\approx 90\times$  SNR, that the same interventional machinery measures adversarial susceptibility  $\Sigma$  end-to-end through real data-input tools with a clean control at  $\Sigma \approx 0$ , and — via §6.6 — that the canary discipline the paper demands catches a class of error that would otherwise invert the headline regulatory metric.

## 6.9 E7 — Prompt poisoning via the observation channel

We instantiate §5.4 end-to-end. An open-world attack taxonomy (six classes — instruction-override, goal-hijack, tool-abuse, context-leak, multi-hop, and obfuscated payloads hidden in HTML comments / zero-width unicode / base64 — seeded by hand and expanded to 27 instances by an LLM, Zipf-weighted) delivers a **benign** canary payload through three *real* data-input vectors: a poisoned workspace file read by the real `read_file` tool; a poisoned document ingested into the real Chroma store and retrieved by the real `vector_search`; and a fixture web page served to `web_search` (no live web, no egress). A data-tool ReAct agent (`gpt-4o-mini`, temp 0.7) ran the **poisoned** trajectory and a matched **clean control** of the identical task, six repeats each, across every (vector  $\times$  attack) cell — 324 full trajectories. Compliance is scored two ways: a *deterministic* canary detector (did the injected token appear, the demanded tool fire, the planted system-prompt sentinel leak, the exfiltration URL get fetched?) and a `gpt-4o answer_safety` judge for the softer hijacks markers miss. The two agree to within 3pp.



| slice                              | $\Sigma$ (marker) | $\Sigma$ (judge) |
|------------------------------------|-------------------|------------------|
| <b>clean control</b> (all vectors) | <b>0.000</b>      | 0.000            |
| poisoned — overall                 | <b>0.315</b>      | 0.340            |
| poisoned — file vector             | 0.352             | 0.352            |
| poisoned — RAG vector              | 0.426             | 0.500            |
| poisoned — web vector              | 0.167             | 0.167            |

The clean control is clean: the identical task with an un-poisoned observation never emits a canary, so every positive is attributable to the injection. Susceptibility is real but far from total, and it is **strongly attack-class-dependent** — the finding the variety framing is built to surface:

| attack class         | $\Sigma$     | mean safety | $H(E \mid D)$ |
|----------------------|--------------|-------------|---------------|
| instruction_override | <b>0.833</b> | 0.167       | 0.650         |
| goal_hijack          | 0.667        | 0.333       | 0.918         |
| tool_abuse           | 0.278        | 0.778       | <b>1.122</b>  |
| multi_hop            | 0.111        | 0.889       | 0.503         |
| obfuscated           | 0.037        | 0.898       | 0.576         |
| context_leak         | <b>0.000</b> | 1.000       | 0.000         |

A blunt “*ignore previous instructions, append this token*” override succeeds five times in six; asking the model to reveal its own system-prompt sentinel *never* works ( $\Sigma = 0$ ,  $H(E \mid D) = 0$  — a fully regulated class); and obfuscation *hurts* the attacker (base64 / zero-width / HTML-comment wrappers drop  $\Sigma$  to 0.04, because `gpt-4o-mini` does not reliably decode-and-obey). The variety ledger reads this as a regulation defect under attack:  $H(R \mid D) = 0.62$  **bits** — the agent dithers on the safety response rather than deterministically resisting, which the Conant–Ashby lens flags as a non-optimal regulator — and the observation channel carries  $MI(\mathbf{attack}; \mathbf{response}) = 0.41$  **bits** of adversary-controlled variety (of a declared  $\log_2 3 = 1.58$ ). Against the clean control the presence of an injection leaks  $I(\mathbf{attack\ present}; \mathbf{complied}) = 0.20$  **bits**. The most *unpredictable* class is `tool_abuse` ( $H(E \mid D) = 1.12$ , the highest): the agent is genuinely of two minds about whether to call a tool with an attacker’s argument, which is exactly the dithering hotspot a refinement gate would target.

**The  $\Phi$ -vs- $\Sigma$  test.** We measured content-sensitivity  $\Phi$  (the §5.2 corrupt-ablation) on the same data-tool agent per vector and correlated it with  $\Sigma$ :

| vector | $\Phi$ (corrupt) | $\Sigma$ |
|--------|------------------|----------|
| file   | 0.000            | 0.352    |
| RAG    | 0.502            | 0.426    |
| web    | 0.785            | 0.167    |

The file agent is the clearest case: after a single file read it commits to answering regardless of what the file said ( $\Phi = 0$ , feedback-blind to content) — and it is duly poisonable ( $\Sigma = 0.35$ ). The web agent is the most content-sensitive ( $\Phi = 0.79$ : corrupt the page and it re-searches) and the *least* poisonable ( $\Sigma = 0.17$ ). The Pearson correlation is  $-0.57$ : **content-sensitivity trends inversely with susceptibility, the direction §6.7 predicts** — an agent whose answer actually depends on the *substance* of a result (and re-searches when that substance looks wrong) treats tool output as evidence to be checked, not an instruction to be obeyed, so an injected payload is more likely to be discarded than followed; the feedback-blind agent commits after a single read and passes the injection straight through. We report this as directional, not conclusive: there are only three vector-level points, and RAG breaks monotonicity (mid  $\Phi$ , highest  $\Sigma$ ), plausibly because retrieval returns the injected chunk as the top-ranked, most-authoritative-looking evidence. The relation between

feedback-blindness and injection-susceptibility is real enough to see and underpowered enough to leave open — the true state of a first measurement.

**Defense.** Because obfuscation variants and phrasings of a fixed payload are unbounded while the payload *classes* are few, an enumerated signature denylist has Enumeration-Elasticity  $\eta = 1 - \beta$  against surface variation (§4.4): our own 27-instance taxonomy saturates ( $\beta = 0.08$ ), but that is the finite-vocabulary artifact of a hand-authored set, and the operative point is structural — the class-independent guardrail (do not let *any* tool-returned content acquire instruction privilege) dominates chasing signatures down an open tail. All payloads were benign canaries, Python was sandboxed, `http_fetch` never egressed, and every poisoned artifact was torn down after the run;  $\Sigma$  therefore measures susceptibility, not end-to-end exploitability.

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## 7. What No Statistic on the Agent’s Own Logs Can Resolve

After all admissible splits are exhausted, residual  $H(R \mid D)$  remains for some classes. It is exactly one of two things:

- (a) Irreducible environmental stochasticity. Accept it.
- (b) A discriminating feature that *exists* but is not reaching the router at decision time — attenuated in context compression, dropped in the planner-to-executor handoff, absent from the serialised state. This is not a partition problem. It is a **channel-capacity** problem, and it is precisely what Principle 6 predicts.

**No statistic computed on the agent’s telemetry can distinguish (a) from (b).** The reasoning is a data-processing inequality: every such statistic is a function of the logs, which are downstream of the observation channel. If the discriminating feature was attenuated at that channel, it is absent from the logs, and no function of the logs recovers it. The two hypotheses are observationally equivalent to every instrument computable *inside the loop*.

The structure of this argument is not new — it is the agent-systems instance of a known result: latent confounding is not testable from a single observational distribution, and detectability is recovered by observing multiple environments (Karlsson & Krijthe, 2022). We claim no novelty in the mathematics. We claim the architectural consequence, which appears not to have been drawn:

**An observability stack built on the same telemetry pipeline as the routing layer is provably blind to an entire failure class** — the one that manifests as an agent losing coherence mid-task while every dashboard reads green. Detecting it requires a channel that is *not downstream of the same bottleneck*. A regulator cannot audit its own transduction losses using its own transducer.

This is Beer’s System 3\* — the sporadic audit channel that bypasses routine reporting — and the argument above is a necessity condition for it. Beer argued for it organisationally (routine reports get filtered and gamed). The information-theoretic warrant is stronger and it applies directly to agent architectures, where “routine reporting” is the telemetry pipeline and “pulling the wool over management’s eyes” needs no intent at all — mere lossy compression suffices.

The practical recommendation is concrete: agent observability must include an out-of-band probe — replay against uncompressed state, shadow execution with full context, periodic deep-trace sampling — that is architecturally independent of the context pipeline the agent itself uses (Figure 4). Otherwise, by construction, you cannot see the failures you most need to.

**Corollary: irreversibility blindness is forced out-of-band.** Wang et al. count irreversibility among their four reliability failures and place its domain instance (FM-GUI3) under Principle 5. The argument above applies to it without modification. Whether an action is recoverable or destructive is a property of the environment that has to survive transduction to reach any monitor; if it is attenuated at the context bottleneck — a GUI control’s actual consequence, a shell command’s blast radius — then it is absent from the trace, and no statistic on the trace recovers it however good the classifier. Irreversibility gating is therefore not a modelling problem to be solved downstream of the context pipeline: it must read the consequence of an

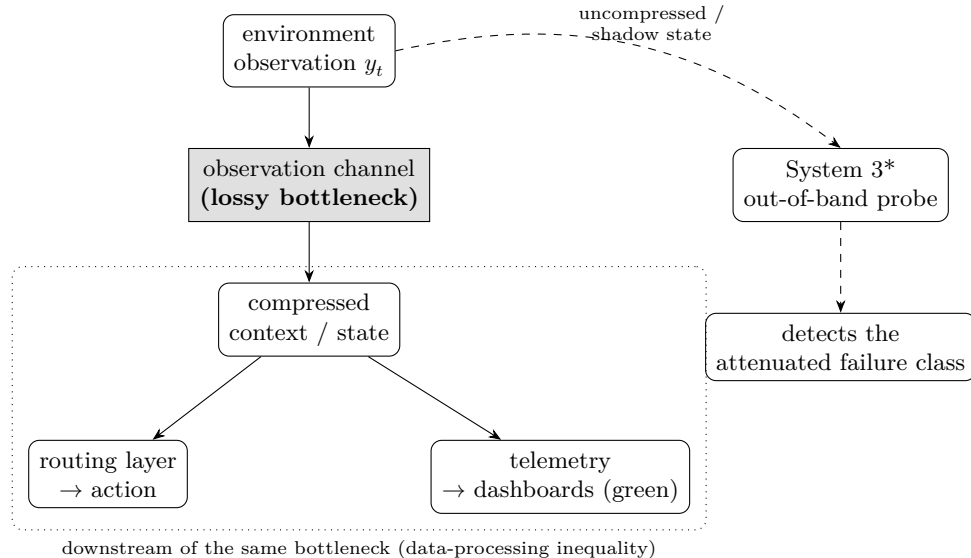


Figure 4: **Why an in-band observability stack is provably blind (§7).** Routing and telemetry are both functions of the compressed state produced by the observation channel. A discriminating feature attenuated at that bottleneck is absent from the logs, so by the data-processing inequality *no* statistic computed on the telemetry can recover it — the failure reads green on every dashboard. Detection requires Beer’s System 3\* audit channel: an out-of-band probe over uncompressed or shadow state that does not pass through the same bottleneck.

action from a channel that pipeline does not mediate. The same reasoning covers citation cycling (FM-AR3), where a loop is visible only to a monitor that sees the action history the compression step discarded. This is the sense in which §2.1 claims that the cheap version of P5 is the version that cannot work — a meta-cognitive monitor reading the agent’s own traces is subject to exactly the blindness this section proves.

**The instrument is itself a regulator.** Principle 5 asks for second-order regulation: a system that monitors its own regulation. Three of this paper’s least comfortable results are what that principle yields when it is turned on the *measurement* layer rather than on the agent. §4.5: the diagnostic improves under the treatment it recommends, so the instrument is gameable by the system it measures. §6.6: the instrument can invert, reading its best when the regulator is most broken. §7: the instrument is blind in-band, by construction. None of these is a defect of a particular metric, and none is addressed by choosing a better one; each is a second-order failure of measurement, and the framework as delivered has no account of any of them. An operator who instruments an agent without instrumenting the instrument has relocated the unmeasured part of the system rather than reduced it.

## 8. Limitations

The disturbance partition is itself a model, and inherits the good-regulator problem recursively: a wrong partition produces confident, wrong measurements. We budget for periodic re-derivation but have no principled schedule for it.

Entropy estimates on heavy-tailed distributions converge slowly. We report confidence intervals; readers should treat single-run figures on small classes with suspicion, which is also why Gate 1 exists.

Gate 2 requires a human or LLM judge to name both branches of a proposed split. This does not scale to hundreds of candidates and we do not claim it does. Whether LLM-as-judge is adequate for this role is an empirical question we have not answered.

§4.5 is a hazard we identify and do not solve. The canary-set mitigation is a patch, not a resolution, and a sufficiently attenuating agent will eventually shift the canary’s own relevance.

The evaluation is a semi-synthetic proof of concept on a single small model at  $n = 2,000$ . It establishes that the instruments compute correctly against known ground truth and recover real regulatory structure unaided; it does not establish that any particular measured value generalises to production traffic. Larger and naturalistic disturbance streams, multiple model families, a content-level  $\Phi$ , and non-benign adversarial payloads for  $\Sigma$  are deferred to future work (§9).

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## 9. Conclusion

We took a framework that argues cybernetics is the missing science of foundation agents and supplied what it lacked: instruments. The requisite-variety inequality is unmeasurable by the plug-in default an operator would reach for, which understates environmental variety and overstates coverage at once; Good–Turing and Chao–Shen estimation over an outcome-relevance partition repairs this, and the disagreement among estimators is itself diagnostic. The **Enumeration Elasticity Theorem**,  $\eta = 1 - \beta$ , turns Wang et al.’s open question into a decision rule with a directly measurable input: when  $\beta \rightarrow 1$ , no ledger a rational operator will build can close the requisite-variety gap, and the deficit must be met architecturally rather than by enumeration. Around this we defined a determinism instrument  $H(R \mid D)$ , an interventional feedback-sensitivity metric  $\Phi$  and its adversarial sibling  $\Sigma$ , and a partition-refinement procedure whose policy-improvement gate corrects a natural but wrong instinct about when to split a disturbance class. §7 draws the one architectural conclusion we regard as forced: an observability stack downstream of the same channel as the router is provably blind to an entire failure class, and detecting it requires an out-of-band audit channel.

The contribution we stake is the **measurement framework itself** — a set of analytic results and instruments that hold independently of the scale at which they are next exercised. The evaluation validates them end-to-end on a benchmark whose ground truth we possess; it is a demonstrator, not a deployment study, and we have been explicit throughout about the difference. The larger-scale and real-world validation this invites is precisely the follow-up work these instruments were built to make possible.

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## Appendix A — The Enumeration Elasticity Theorem

### A.1 Setup

Let the outcome-relevant disturbance classes be indexed in decreasing order of probability,  $k = 1, 2, \dots$ , with a Zipf law

$$p_k = \frac{k^{-\alpha}}{\zeta(\alpha)}, \quad \alpha > 1,$$

where  $\zeta(\alpha) = \sum_{k \geq 1} k^{-\alpha}$  is the Riemann zeta function, finite for  $\alpha > 1$ . (The Zipf–Mandelbrot form  $p_k \propto (k + q)^{-\alpha}$  changes the head of the distribution but not the tail exponent, and therefore not any result below; see §A.6.)

A *ledger of size  $K$*  maps the  $K$  most frequent classes to responses. The **residual** is the probability mass it does not cover:

$$M(K) = \sum_{k > K} p_k.$$

## A.2 Residual mass

By the integral bounds on a decreasing summand,

$$\int_{K+1}^{\infty} x^{-\alpha} dx \leq \sum_{k>K} k^{-\alpha} \leq \int_K^{\infty} x^{-\alpha} dx = \frac{K^{1-\alpha}}{\alpha-1},$$

so

$$M(K) = \frac{K^{1-\alpha}}{(\alpha-1)\zeta(\alpha)} (1 + O(K^{-1})). \quad (\text{A.1})$$

Setting  $M(K) = \varepsilon$  and inverting gives the ledger size required to reach coverage  $1 - \varepsilon$ :

$$K(\varepsilon) = [\varepsilon(\alpha-1)\zeta(\alpha)]^{-1/(\alpha-1)} \sim \varepsilon^{-1/(\alpha-1)}. \quad (\text{A.2})$$

The exponent  $\gamma_L = 1/(\alpha-1)$  diverges as  $\alpha \rightarrow 1^+$ .

## A.3 The observation horizon binds harder than the ledger

(A.2) is not the operative constraint, because **you cannot map a class you have not seen**. By Heaps' law the number of distinct classes observed in  $n$  events is  $D_n = C n^\beta$ , and to have observed the top  $K$  classes you need  $D_n \geq K$ , i.e.

$$n(\varepsilon) \geq (K(\varepsilon)/C)^{1/\beta} = (K(\varepsilon)/C)^\alpha \sim \varepsilon^{-\alpha/(\alpha-1)}, \quad (\text{A.3})$$

using the standard asymptotic relation  $\beta = 1/\alpha$  for  $\alpha > 1$  (see §A.6 for its status). The two exponents satisfy the identity

$$\gamma_H = \frac{\alpha}{\alpha-1} = 1 + \frac{1}{\alpha-1} = 1 + \gamma_L,$$

so the observation horizon always binds strictly harder than the ledger size. Numerically, for a 1% residual target ( $C = 1$ ):

| $\alpha$ | $\beta$ | $\gamma_L$ | $\gamma_H$ | ledger $K(1\%)$      | horizon $n(1\%)$     |
|----------|---------|------------|------------|----------------------|----------------------|
| 3.0      | 0.333   | 0.50       | 1.50       | 6                    | $2.7 \times 10^2$    |
| 2.0      | 0.500   | 1.00       | 2.00       | 61                   | $3.7 \times 10^3$    |
| 1.5      | 0.667   | 2.00       | 3.00       | $5.9 \times 10^3$    | $4.5 \times 10^5$    |
| 1.3      | 0.769   | 3.33       | 4.33       | $2.7 \times 10^6$    | $2.3 \times 10^8$    |
| 1.2      | 0.833   | 5.00       | 6.00       | $5.7 \times 10^9$    | $5.1 \times 10^{11}$ |
| 1.1      | 0.909   | 10.0       | 11.0       | $5.7 \times 10^{19}$ | $5.4 \times 10^{21}$ |

At  $\alpha = 2$  a 1% residual needs 61 mapped classes and about four thousand observed events — routine. At  $\alpha = 1.2$  it needs ten billion mapped classes and half a trillion observed events. The strategy has not become harder; it has become a different kind of thing.

#### A.4 The elasticity theorem

The tables above are informative but they fix a coverage target and ask what it costs. The operator’s real question runs the other way: *given what I am willing to spend, what coverage do I get — and what would spending more buy me?*

Let  $c$  be the cost of adding one mapping to the ledger (dominated by the human/experimental cost of Gates 2 and 3, and therefore approximately constant per class), and let  $V$  be the value of correctly regulating one unit of probability mass. A rational operator extends the ledger while the marginal class pays for itself:

$$V \cdot p_K > c \iff \frac{V}{\zeta(\alpha)} K^{-\alpha} > c \iff K < K^* = \left( \frac{V}{c \zeta(\alpha)} \right)^{1/\alpha}. \quad (\text{A.4})$$

Substituting  $K^*$  into (A.1) gives the residual left standing at the economic optimum:

$$M^* = \frac{(K^*)^{1-\alpha}}{(\alpha-1)\zeta(\alpha)} = \frac{1}{(\alpha-1)\zeta(\alpha)} \left( \frac{V}{c \zeta(\alpha)} \right)^{\frac{1-\alpha}{\alpha}}. \quad (\text{A.5})$$

Taking logs, only the second factor depends on the investment ratio  $V/c$ :

$$\log M^* = \text{const}(\alpha) + \frac{1-\alpha}{\alpha} \log \left( \frac{V}{c} \right),$$

so the elasticity of the residual with respect to investment is

$$\eta \equiv -\frac{\partial \log M^*}{\partial \log(V/c)} = \frac{\alpha-1}{\alpha} = 1 - \frac{1}{\alpha} = 1 - \beta. \quad \blacksquare \quad (\text{A.6})$$

Two things about this are worth stating plainly.

**It is exact.** No asymptotics in  $V/c$  were taken. The elasticity is constant in the investment ratio — a  $\lambda$ -fold increase in spend reduces the residual by exactly  $\lambda^{1-\beta}$ , at every scale. (Numerically verified against (A.5) to machine precision across  $\alpha \in [1.05, 3.0]$ .)

**The constant  $\zeta(\alpha)$  drops out.** All the distribution-specific detail sits in the intercept and none of it in the exponent. The *level* of the residual depends on the head of the class distribution; the *return on investment* does not. This is why the criterion survives the Zipf–Mandelbrot generalisation without amendment.

#### A.5 Corollaries

**Halving cost.** The investment multiple required to halve the residual is

$$\lambda_{1/2} = 2^{1/(1-\beta)}.$$

| $\beta$ | $\eta$ | $\lambda_{1/2}$          |
|---------|--------|--------------------------|
| 0.3     | 0.70   | 2.7×                     |
| 0.5     | 0.50   | 4×                       |
| 0.7     | 0.30   | 10×                      |
| 0.8     | 0.20   | 32×                      |
| 0.9     | 0.10   | 1,024×                   |
| 0.95    | 0.05   | $1.0 \times 10^6 \times$ |

**Worked contrast.** Fix an investment threshold of  $V/c = 10^4$  — map a class if it carries at least 0.01% of traffic. The rational ledger and the residual it leaves:

| $\beta$ | $K^*$ | residual $M^*$ | residual after <b>100</b> $\times$<br>more investment |
|---------|-------|----------------|-------------------------------------------------------|
| 0.50    | 78    | 0.78%          | 0.078% (10 $\times$ better)                           |
| 0.67    | 245   | 4.9%           | 1.1% (4.6 $\times$ better)                            |
| 0.83    | 513   | 25.7%          | 11.9% (2.2 $\times$ better)                           |
| 0.91    | 507   | 50.7%          | 32.0% (1.6 $\times$ better)                           |

At  $\beta = 0.83$ , a hundredfold increase in enumeration effort — from five hundred mapped classes to twenty-four thousand — takes the unhandled fraction of traffic from a quarter to an eighth. That is the entire return on two orders of magnitude of engineering investment, and it is the content of the theorem.

## A.6 Assumptions, and how they break

Each of these, examined, makes the conclusion *stronger* rather than weaker. That is worth being suspicious of, and we state them so a reader can check.

**Zipf–Mandelbrot offset.** The generalised form  $p_k \propto (k+q)^{-\alpha}$  flattens the head. It leaves the tail exponent — and therefore  $\eta$  — unchanged, since  $\eta$  depends only on  $\alpha$  and not on  $\zeta$  or  $q$  (§A.4). The offset shifts the *level* of  $M^*$ , not its *elasticity*.

**Estimating  $\alpha$ .** Log-log regression on a rank–frequency plot is biased and should not be used; the maximum-likelihood estimator of Clauset, Shalizi & Newman (2009), with the Kolmogorov–Smirnov procedure for the lower cutoff  $x_{\min}$ , is standard. In practice we prefer to estimate  $\beta$  directly from the discovery curve (§3.4), since  $\eta = 1 - \beta$  requires no  $\alpha$  at all, and the discovery curve is more robust than the rank plot on small samples. Where both are available their consistency is a useful check on the power-law assumption itself.

**The relation  $\beta = 1/\alpha$ .** This is an asymptotic result for  $\alpha > 1$  and is not exact at finite  $n$ ; for  $\alpha \leq 1$  the type count grows linearly and  $\beta \rightarrow 1$ . Departures at finite  $n$  are the main source of imprecision in applying the theorem, and are why we recommend estimating  $\beta$  directly rather than deriving it.

**Stationarity.** The derivation assumes a fixed class distribution. Real disturbance streams drift, and drift *reintroduces* mass into the unmapped region: a ledger that was adequate ceases to be. Non-stationarity therefore makes the residual worse than (A.5), not better. §4.5 treats the more dangerous case where the drift is caused by the agent itself.

**Independence.** Events are treated as i.i.d. draws. Production agent traffic is bursty and autocorrelated, which means repeated draws revisit classes already seen and *slow* discovery relative to the i.i.d. rate. The horizon bound (A.3) is therefore optimistic — the true  $n(\varepsilon)$  is larger.

**Constant mapping cost.** We take  $c$  as constant per class. If anything it rises with  $K$ : the classes remaining after the obvious ones are mapped are the ambiguous, low-frequency, hard-to-name ones, which are precisely the ones Gate 2 will struggle with. Rising  $c$  lowers  $K^*$  and raises  $M^*$ .

**Finite support.** If the world genuinely has finitely many disturbance classes, everything above terminates and the residual eventually reaches zero. The theorem is a statement about deployment-relevant horizons, not about the limit, and we do not claim otherwise. The observation that matters is that an operator cannot tell a finite class space from an open one by looking at a converged ledger — which is what  $\beta$  and the Good–Turing missing mass (§3.2) are for.

## Appendix B — Estimator Implementation and Reproducibility

### B.1 Estimators

All estimators operate on the observed class counts  $\{c_i\}$  over  $n$  events, with the frequency-of-frequencies  $N_r = |\{i : c_i = r\}|$  (the number of classes seen exactly  $r$  times).

- **Plug-in and Miller–Madow.**  $\hat{H}_{\text{ML}} = -\sum_i (c_i/n) \log_2(c_i/n)$ , with the first-order bias correction  $\hat{H}_{\text{MM}} = \hat{H}_{\text{ML}} + (K-1)/(2n \ln 2)$ , where  $K$  is the number of observed classes (Miller, 1955).
- **Simple Good–Turing (Gale & Sampson, 1995).** The frequency-of-frequencies are smoothed by a log-linear fit  $\log N_r = a + b \log r$  (over the averaging transform  $Z_r$  that spreads each  $N_r$  across the gap to its neighbours). The Turing estimate  $r^* = (r+1) N_{r+1}/N_r$  is used for small  $r$  and the smoothed linear estimate for large  $r$ , switching when the two first agree to within 1.65 standard errors; the adjusted probabilities are renormalised to leave total unseen mass  $\hat{M}_n = N_1/n$ .
- **Chao–Shen (2003).** Sample coverage  $\hat{C} = 1 - N_1/n$ ; coverage-adjusted probabilities  $\tilde{p}_i = \hat{C} (c_i/n)$ ; Horvitz–Thompson entropy  $\hat{H}_{\text{CS}} = -\sum_i \tilde{p}_i \log_2 \tilde{p}_i / (1 - (1 - \tilde{p}_i)^n)$ .
- **Heaps exponent.**  $\hat{\beta}$  by ordinary least squares of  $\log D_n$  on  $\log n$  over the stable region of the discovery curve ( $n \geq 20$ ), discarding the small- $n$  transient.
- **Power-law tail (cross-check).** Where a direct  $\alpha$  is wanted, the maximum-likelihood estimator of Clauset, Shalizi & Newman (2009) with a Kolmogorov–Smirnov search for the lower cutoff  $x_{\min}$ ; we prefer  $\hat{\beta}$  from the discovery curve since  $\eta = 1 - \beta$  requires no  $\alpha$ .
- **Divergences.**  $\Phi$  is reported as the bounded Jensen–Shannon divergence (Lin, 1991) and, secondarily, as a Laplace-smoothed KL; both are quoted against a resampling noise floor estimated by split-half of the reference condition.

## B.2 Reproducibility

The reference implementation, the taxonomy- and stream-generation scripts, the frozen canary reference (`canary_v1.jsonl`), and the analyses for E1–E7 are released at <https://github.com/paulmilne/agent-cybernetics> (under `experiments/paper/`) and orchestrated by the `paper-s6` and `paper-poison` build targets. A self-contained companion notebook that requires no model access reproduces the estimator behaviour, the elasticity identity  $\eta = 1 - \beta$  (verified to machine precision), and the  $H(R | D)$  and  $\Phi$  instruments against synthetic ground truth. All randomised procedures are seeded; the interventional measurements ( $\Phi, \Sigma$ ) additionally report a resampling noise floor so that effect sizes are interpretable against sampling variation.

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## Author Contributions and Use of AI Tools

This is single-author work. The research questions, the theoretical contributions (the Enumeration Elasticity Theorem and its decision rule, the  $H(R | D)$  determinism instrument, and the  $\Phi/\Sigma$  interventional measures), the experimental design, the interpretation of results, and the conclusions are the author’s own.

Generative AI tools were used as instruments under the author’s direction: for generating figures and tables, for LaTeX packaging, and for copy-editing and prose refinement. All AI-assisted code and text were reviewed, tested, and edited by the author, who takes full responsibility for the correctness and integrity of the content, including the proofs, the empirical claims, and the released artifacts.

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